

Muhammad Shayan Nazeer

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RESEARCH PROFILE

Ph.D. researcher in networks and distributed systems building low-latency infrastructure for mobile and edge AI. My work spans programmable 5G/6G core networks, distributed state consistency, fault-tolerant control planes, RAN mobility, KV-cache optimization, and test-time scaling for LLM serving at the mobile edge.

EDUCATION

PhD, Electrical and Computer Engineering

University of Massachusetts Amherst, USA

Sep. 2023 – Dec 2027 (Expected)

Advisor: [Dr. Taqi Raza](#)

- Research focus: networks and distributed systems for programmable mobile infrastructure, distributed state consistency, low-latency orchestration, and AI serving at the edge.

Bachelor's in Electrical Engineering

National University of Sciences and Technology (NUST), Islamabad, Pakistan

Sep. 2019 – Jul. 2023

Gold Medal

- Focus: wireless networks, embedded systems, machine learning, signal processing, computer architecture, and system security.

RESEARCH EXPERIENCE

University of Massachusetts Amherst

Graduate Research Assistant

Amherst, MA, USA

Sep. 2023 – Present

- Built CoreX, a distributed 5G/6G core that executes mobile-control procedures as DAG-scheduled workers over consistency-aware shared state.
- Reduced control-plane latency by up to 2.1x for registration, 10.4x for paging, and 16.6x for N2 handover; improved failure-recovery p99 latency from 622 ms to 17 ms.
- Designing semantic KV-cache management for long-context LLM inference at the edge, where GPU memory, network bandwidth, and latency budgets constrain serving decisions.
- Exploring topic-aware KV-cache organization, eviction, merging, and precision adaptation to reduce memory pressure while preserving reasoning quality.
- Studying test-time scaling policies that allocate compute dynamically across prompts, models, and edge/cloud execution paths based on latency, uncertainty, and task difficulty.

Nokia Bell Labs

Research Scientist Intern

Murray Hill, NJ, USA

Feb. 2025 – Jun. 2025

- Built an reinforcement learning mobility framework that uses lower-layer radio measurements to prepare target cells before handover execution, reducing late synchronization in 5G NR mobility procedures.
- Implemented Lower Layer Triggered Mobility end-to-end across 3 protocol layers: MAC, RRC, and control-plane signaling.
- Designed a RACH-less handover protocol that removes target-cell random access signaling and reduces mobility overhead by 25
- Co-invented a Nokia Bell Labs patent on early synchronization and RACH-less mobility for 5G/6G RAN systems.

NUST NeuroInformatics Research Lab

Machine Learning Engineer

Islamabad, Pakistan

Sep. 2022 – Aug. 2023

- Built a real-time EEG inference pipeline for motor-intention decoding, integrating 4 stages: signal acquisition, preprocessing, deep-learning classification, and robotic actuation.
- Integrated the inference stack with an assistive robotic control system under real-time latency constraints; work published at IEEE IC2AI 2025.

PATENTS

Method for Early Synchronization Through Source Cell for RACH-Less LTM

Muhammad Shayan Nazeer, A. Kak, N. Choi

Accepted, Nokia Patents Board

PUBLICATIONS

CoreX: Next-G Cellular Core Network for Emerging Applications <i>Muhammad Shayan Nazeer, MT Raza</i>	under review
FedAir: Network-Aware Federated Learning over 5G <i>MA Soomro, Muhammad Shayan Nazeer, FM Anwar, MT Raza</i>	<i>ACM MobiCom 2026, under review</i>
Characterizing Cellular Network Dynamics for Federated Human Activity Recognition <i>MA Soomro, Muhammad Shayan Nazeer, MT Raza, FM Anwar</i>	<i>FmSys @ ACM SenSys 2026</i>
SynchroNB: Toward Robust Timing for 5G NB-IoT Networks <i>MA Soomro*, Muhammad Shayan Nazeer*, C. Delsignore, Y. Chandio, MT Raza, FM Anwar</i>	<i>ACM SenSys 2026</i>
ORACLE: Reconciling Next-G Cellular Core Operations for Correctness <i>Muhammad Shayan Nazeer, MT Raza</i>	<i>IEEE ICC 2026</i>
Non-Invasive EEG-Driven Brain-Computer Interface for Motion Control of Assistive Robotics <i>S. E. Fatima, S. A. Rahman, W. Mumtaz, Muhammad Shayan Nazeer</i>	<i>IEEE IC2AI 2025</i>

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Bash, JavaScript, MATLAB
- **Systems:** Distributed Systems, Fault Tolerance, State Consistency, Scheduling, Linux, Docker, Kubernetes, Git
- **Networking:** 5G/6G Core Networks, RAN Mobility, Control-Plane Design, SDN, NFV, Traffic Engineering, Protocol Analysis
- **AI Infrastructure:** LLM Serving, KV-Cache Optimization, Test-Time Scaling, Edge Inference, Model Serving, Device-Edge-Cloud Offloading
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Reinforcement Learning, Federated Learning, RAG, Pandas, NumPy

HONORS & AWARDS

NSF CONEXT Travel Grant Awarded travel support to attend ACM CoNEXT.	2025
Colosseum Young Gladiator NSF-funded master class on the Colosseum 5G testbed, Northeastern University.	2024
Rector's Gold Medal Awarded for best undergraduate senior design project, NUST.	2023
NUST High Achiever's Award Gold Medal for outstanding academic performance.	2023
CSAW Logic Locking Conquest Global Runner Up, organized by NYU.	2022
National Physics Talent Contest Selected among top 50 students in Pakistan for IPhO training.	2018
Intel ISEF National-level finalist.	2017

REFERENCES

Dr. Muhammad Taqi Raza — Ph.D. Advisor
Assistant Professor, ECE, University of Massachusetts Amherst
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Dr. Nak Jung Choi — Research Internship Supervisor
Department Head, Mobile Network Systems, Nokia Bell Labs, Murray Hill, NJ
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Dr. Fatima M. Anwar — Research Collaborator
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